

1. Ecology is the scientific study of... interactions among organisms, populations, and communities, of organisms effects on their environment, and their responses to changes in their environment.
2. Making Connections: How are Ecology and Economics connected? Both Ecology & Economics are the studies of "houses". Ecology = Nature ; Economics = Humans
Economics interacts w/ Ecology because we share 1 house = BIOSPHERE
3. Why do we need to understand ecology? so that we can design human economies that are sustainable - which means that they function without degrading the environment.

The kinds of questions that ecologists may ask about the living environment can vary, depending on the level at which the ecologist works. In the table below; write the levels of organization of the biosphere (from smallest to largest) AND **DEFINE** each level

Level (smallest to largest)	Definition/ what it includes
Species	a group of similar organisms that breed & produce fertile offspring
Population	a group of individuals that belong to the same species & <u>live in the same area</u>
Community	Different populations that live together in a defined area
Ecosystem	<u>ALL</u> organisms that live in a place, <u>together with</u> their physical environment
BIOME	A group of ecosystems that <u>share similar climates & typical organisms</u>
Biosphere	Our entire planet, with all its organisms & physical environments

4. Modeling: Create a graphic organizer below that shows the relationships of the levels of organization defined above.

